The two T^2 gerbe ends are coupled, not independent: a strict one-way forced correlation $\delta \Rightarrow [\mathfrak{g}]$

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Abstract

For a structure group K of nilpotency class ≤ 2, the nonabelian inventory gerbe on the torus $T^2 = \mathrm{B}\mathbb{Z}^2$ carries two obstruction classes sharing the hinge $\mathrm{Inn}K$: the *strictification* class $\delta(\rho) = [A, B] \in H^2(T^2; \mathrm{Inn}K)$ and the *central* class $[\mathfrak{g}] \in H^2(T^2; Z(K))$. We prove they are *not* independent. They satisfy a forced one-way correlation

$$\delta \text{ couples on } T^2 \implies [\mathfrak{g}] \text{ couples on } T^2,$$

so the corner ($\delta \neq 0$, $[\mathfrak{g}] = 0$) is *forced empty*. The cause is a structural asymmetry between two abelian-kernel extensions: the central extension $1 \to Z(K) \to K \to \mathrm{Inn}K \to 1$ is non-split for every nonabelian class-2 K and carries the *trivial* $\mathrm{Inn}K$-action, so $[\mathfrak{g}]$ universally couples, while the strictification extension $\mathrm{Aut}K \to \mathrm{Out}K$ *can* split, so δ is selective. The implication is *strict*: the converse fails at $K = D_4$ ($[\mathfrak{g}] \neq 0$ while $\delta \equiv 0$), the dihedral $4 \mid n$ witness at $n = 4$. This is the T^2-analogue of the S^1-floor $D1 \Rightarrow D2$ dichotomy. A battery cross-tabulation confirms the empty corner.

1 Verdict

Over the base $T^2 = \mathrm{B}\mathbb{Z}^2$ we decide, for the two Giraud obstruction stages of the inventory gerbe, between

- **(A)** *independence* — both off-diagonal corners ($\delta \neq 0, [\mathfrak{g}] = 0$) and ($\delta = 0, [\mathfrak{g}] \neq 0$) realized; and
- **(B)** *forced correlation* — some implication always holds.

The answer is **(B)**. For every finite group K of nilpotency class ≤ 2 (the exact domain on which both obstructions are abelian-H^2-valued),

$$\boxed{\delta \text{ couples on } T^2 \implies [\mathfrak{g}] \text{ couples on } T^2}$$

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i.e. the corner ($\delta \neq 0,\ [\mathfrak{g}] = 0$) is *forced empty*. The converse corner ($\delta = 0,\ [\mathfrak{g}] \neq 0$) *is* occupied (e.g. $K = D_4$, the dihedral $4 \mid n$ witness at $n = 4$), so the correlation is *strict and one-directional*. There is no logical independence.

The group-theoretic cause is a structural asymmetry between the two extensions that the symmetric picture — "both are commutator obstructions of an abelian-kernel extension through the hinge Inn$K$" — hides:

- the *central* extension $1 \to Z(K) \to K \to \mathrm{Inn}K \to 1$ is *non-split for every nonabelian* class-2 K, and its coefficient $Z(K)$ carries the *trivial* Inn$K$-action, so $[\mathfrak{g}]$ is *universally obstructed* — every nonabelian class-2 group couples at the central end;
- the *strictification* extension $1 \to \mathrm{Inn}K \to \mathrm{Aut}K \to \mathrm{Out}K \to 1$ *can split* (it does for $D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$), killing δ *selectively*.

Strictification coupling is therefore a strictly stronger non-commutativity condition than central coupling, and it cannot occur without the weaker one.

2 Setup: the two ends as one lemma, applied twice

Let K be finite of nilpotency class ≤ 2, so $[K, K] \subseteq Z(K)$ and $M := \mathrm{Inn}K = K/Z(K)$ is *abelian*. Two abelian-kernel group extensions share the hinge Inn$K$:

$$(\Sigma)\quad 1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{\pi} \mathrm{Out}K \to 1, \qquad (Z)\quad 1 \to Z(K) \to K \xrightarrow{\mathrm{ad}} \mathrm{Inn}K \to 1.$$

Over the base $T^2 = \mathrm{B}\mathbb{Z}^2$ (nerve of $\mathcal{C} = \mathrm{B}\mathbb{Z}^2$), the proved δ-commutator lemma (`coupled-T2-general-delta-lemma`) computes the degree-2 connecting/lifting obstruction of *any* such extension $1 \to A \to E \to Q \to 1$ (A abelian) for a band $b : \mathbb{Z}^2 \to Q$: with lifts $x, y \in E$ of $b(s), b(t)$,

$$\mathrm{obs}(b) = \big[\,[x, y]\,\big] \in H^2(T^2; A) = A\big/\big((\sigma{-}1)A{+}(\tau{-}1)A\big), \qquad [x, y] = xyx^{-1}y^{-1} \in \ker(E \to Q) = A,$$

independent of lifts, where σ, τ are the b-twisted actions of s, t on A. We instantiate it on both extensions.

Strictification end (Σ)**.** Band $\rho : \mathbb{Z}^2 \to \mathrm{Out}K$, lifts $A, B \in \mathrm{Aut}K$:

$$\delta(\rho) = \big[[A, B]\big] \in H^2(T^2; \mathrm{Inn}K).$$

Central end (Z)**.** Here the coefficient is $A = Z(K)$ and the action is *trivial*: for $z \in Z(K)$ and any lift $k \in K$ of $\bar{k} \in \mathrm{Inn}K$, $\bar{k} \cdot z = kzk^{-1} = z$ (central). Hence $(\sigma - 1)Z(K) + (\tau - 1)Z(K) = 0$ and

$$H^2(T^2; Z(K)) = Z(K) \quad \text{(trivial coefficients, no quotient)}.$$

An *inner band* $\beta : \mathbb{Z}^2 \to \mathrm{Inn}K$ (automatically valid: Inn$K$ abelian $\Rightarrow$ any pair of images commutes) with lifts $k, k' \in K$ gives

$$[\mathfrak{g}] = \gamma(\beta) = \big[[k, k']\big] = [k, k'] \in Z(K).$$

This is exactly the registry's Face 2 central-extension lifting class (`lift-cocycle-derivation`, proved), now read through the same lemma.

Remark 1 (What "independence" means here)**.** δ and $[\mathfrak{g}]$ live in *different* coefficient groups (Inn$K$ vs $Z(K)$) and arise from *different* band targets (Out$K$ vs Inn$K$); for a single physical gerbe the central stage is only reached *after* strictification succeeds. So the independence question is correctly posed at the level of the *group* K: does K admit a coupling band at each end? — the four-corner cross-tabulation. We decide that.

3 Two lemmas

Lemma 2 (central end is universally obstructed)**.** *For K finite of class ≤ 2: $[\mathfrak{g}] \neq 0$ for some inner band $\iff$ K is nonabelian.*

Proof. ($\Leftarrow$) K nonabelian $\Rightarrow \exists\, k, k' \in K$ with $[k, k'] \neq 1$. By class ≤ 2, $[k, k'] \in [K, K] \subseteq Z(K)$, so $[k, k'] \in Z(K) \setminus \{1\}$. Their images $\bar{k}, \bar{k}' \in \mathrm{Inn}K$ commute (abelian), so $\beta : s \mapsto \bar{k},\ t \mapsto \bar{k}'$ is a band, and by §2 $\gamma(\beta) = [k, k'] \neq 0$ in $H^2(T^2; Z(K)) = Z(K)$. ($\Rightarrow$) If K abelian then $[k, k'] = 1$ for all k, k', so $\gamma(\beta) = 0$ for every band. □

Lemma 3 (strictification end needs a nonabelian group)**.** *For any finite K: $\delta(\rho) \neq 0$ for some band ρ $\Rightarrow$ K is nonabelian.*

Proof. Contrapositive. If K is abelian then $\mathrm{Inn}K = 1$, so (Σ) reads $1 \to 1 \to \mathrm{Aut}K \xrightarrow{=} \mathrm{Out}K \to 1$: the kernel is trivial, every band lifts uniquely, and $\delta \equiv 0$ (the lemma gives $[A, B] \in \mathrm{Inn}K = 1$). □

4 Theorem: forced correlation

Theorem 4 (forced correlation)**.** *Let K be finite of nilpotency class ≤ 2. Then*

$$\left[\ \exists\, \rho : \delta(\rho) \neq 0\ \right] \implies \left[\ \exists\, \beta : [\mathfrak{g}] = \gamma(\beta) \neq 0\ \right].$$

Equivalently, the corner (δ couples, $[\mathfrak{g}] = 0$ for all bands) is empty*. The implication is* strict*: its converse fails — $K = D_4$ has $[\mathfrak{g}] \neq 0$ but $\delta \equiv 0$.*

Proof. If $\delta(\rho) \neq 0$ for some ρ, Lemma 3 gives K nonabelian, and then Lemma 2 gives a band β with $[\mathfrak{g}] = \gamma(\beta) \neq 0$. Strictness: for $K = D_4$ (dihedral of order 8, class 2), $\mathrm{Inn}D_4 = (\mathbb{Z}/2)^2$, and the extension (Σ), $1 \to (\mathbb{Z}/2)^2 \to \mathrm{Aut}D_4 \to \mathrm{Out}D_4 \to 1$ with $\mathrm{Aut}D_4 \cong D_4$, $\mathrm{Out}D_4 \cong \mathbb{Z}/2$, *splits* — the order-2 automorphism rs (outside the inner Klein four-subgroup $\{1, r^2, s, r^2 s\}$ of $\mathrm{Aut}D_4$) is a homomorphic section — so $\delta \equiv 0$. Yet D_4 is nonabelian, so by Lemma 2 $[\mathfrak{g}] = [r, s] = r^2 \neq 0 \in Z(D_4)$. □

The structural cause (the asymmetry). The two "split?" questions underlying the four corners are *not* logically free:

- **(Z) is non-split for every nonabelian class-2 K.** A splitting $\mathrm{Inn}K \to K$ would be a homomorphic section with image a complement S to $Z(K)$; because (Z) is central, $K = Z(K) \times S$ with $Z(K)$ and $S \cong \mathrm{Inn}K$ both abelian, forcing K abelian — a contradiction. (Coupling is then supplied directly by Lemma 2, using the trivial center-action.)
- **(Σ) can split** ($D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$ all do), and split $(\Sigma) \Rightarrow \delta \equiv 0$ (a homomorphic section $\theta : \mathrm{Out}K \to \mathrm{Aut}K$ turns any band ρ into a homomorphic lift $\theta\rho$).

So the column "(Z) splits" of the cross-tabulation contains *only abelian groups*; the central end is forced on for all nonabelian K, while the strictification end is free. That collapse of one column *is* the forced correlation.

5 Computational cross-tabulation (confirmatory)

The script `2026-10-13-two-ends-crosstab.py` computes, for each battery group: $|Z|$, $|\mathrm{Aut}|$, $|\mathrm{Inn}|$, $|\mathrm{Out}|$; whether $\mathrm{Inn}K$ is abelian (class ≤ 2); whether (Σ) splits and whether it δ-couples (exhaustive search for a commuting outer pair with no commuting lift); whether (Z) splits and whether it $[\mathfrak{g}]$-couples.

K	class ≤ 2?	(Σ) split	δ couples	(Z) split	$[\mathfrak{g}]$ couples
D_4	yes	**yes**	no	no	**yes**
Q_8	yes	yes	no	no	yes
$\mathrm{Heis}(\mathbb{F}_3)$	yes	yes	no	no	yes
M_{16}	yes	yes	no	no	yes
$C_4 \rtimes C_4$	yes	yes	no	no	yes
$D_4 \times C_2$	yes	no	**yes (30)**	no	yes
$Q_8 \times C_2$	yes	no	**yes (54)**	no	yes
SD_{16}, D_{16}	**no** (Inn nonab.)	—	no	no	(nonab.)
C_4, $(\mathbb{Z}/2)^2$ (abelian)	yes	yes	no	**yes**	no

Four corners (δ couples) $\times$ ($[\mathfrak{g}]$ couples):

- (yes, yes): $D_4 \times C_2$, $Q_8 \times C_2$.
- (yes, no): **EMPTY** — the forced-correlation corner.
- (no, yes): $D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$ (every split-(Σ) nonabelian class-2 group).
- (no, no): abelian K only (vacuous).

Every nonabelian row has (Z) non-split and $[\mathfrak{g}]$ coupling; every abelian row has (Z) split and no coupling — exactly Lemmas 2–3. The decisive (yes, no) corner is empty, confirming the Theorem. (SD_{16}, D_{16} are included to mark the boundary: Inn is nonabelian there — class 3 — so the δ-end is *outside* the clean ordinary-H^2 domain; they still show no violation, but the theorem is not claimed for them.)

6 Verification (hostile-referee pass)

- **§2 central-end identification.** Trivial center-action is immediate (z central). The lemma applies verbatim to any abelian-kernel extension; (Z) qualifies with $A = Z(K)$. $H^2(T^2; Z(K)) = Z(K)$ for trivial coefficients is standard ($T^2 = K(\mathbb{Z}^2, 1)$, $H^2(\mathbb{Z}^2; M) = M$). The class is the bare commutator — matches `lift-cocycle-derivation` and the D_4 arithmetic $[r, s] = r^2$.
- **Lemma 2.** Class ≤ 2 is used exactly once and essentially: to land $[k, k']$ in $Z(K)$. Without it $[k, k']$ need not be central and the inner-band obstruction is not defined by this formula — this is why the domain is class ≤ 2, not accidental.
- **Lemma 3.** Elementary; $\mathrm{Inn}K = 1 \Rightarrow \delta \equiv 0$.
- **Theorem 4.** Pure composition of the two lemmas; the empty corner is their conjunction. The structural-cause paragraph (central extension of a nonabelian class-2 group never splits) is an independent elementary check, and matches (Z)-split=False on every nonabelian computed row.

- **Strictness witness D_4.** $\mathrm{Aut}D_4 \cong D_4$ splits over the inner Klein four via the order-2 element rs; verified by the engine (Σ split=True, δ-couple=False) and by hand. D_4 is the dihedral $4 \mid n$ instance at $n = 4$.

- **Boundary/degenerate cases.** Abelian K: both ends trivial (corner $(\mathrm{no}, \mathrm{no})$). Rank-1 base S^1: $H^2(S^1; -) = 0$, both ends vanish, no content. Trivial band: commutator $= 0$, both obstructions 0.

- **Scope honesty.** SD_{16}, D_{16} (and dihedral D_n, $4 \mid n$, $n > 4$) have Inn nonabelian: outside the clean domain, explicitly excluded, no claim made.

7 Scope and (non-)gaps

- **Domain = class ≤ 2, and this is the natural maximal domain.** Both obstructions are ordinary abelian H^2 precisely when $\mathrm{Inn}K$ is abelian (class ≤ 2): then $\delta \in H^2(T^2; \mathrm{Inn}K)$ and (Z) is a genuine central extension with $[k, k'] \in Z(K)$. Outside it the δ-end becomes nonabelian-H^2 (Giraud/Eilenberg–MacLane) and the central-end formula needs the commuting-in-Inn caveat; the clean correlation statement is not claimed there and is the one honest open edge (the dihedral D_n, $4 \mid n$, $n \geq 8$ family lives here — computed rows show no violation but the framework is only heuristic).

- **Grade.** The Theorem (forced correlation) is *proved* — it rests only on the proved δ-commutator lemma and two elementary lemmas. The battery cross-tabulation is *computed* over the enumerated family (not a census of all groups of each order; minimality is not claimed).

- **Not inflated.** The claim is a one-directional implication with a strict counterexample to its converse, *not* a biconditional and *not* a census theorem. The decisive content is the forced-empty corner plus its structural cause (the (Z)-split column is abelian-only).

One-line summary. On T^2, the strictification obstruction $\delta \in H^2(T^2; \mathrm{Inn}K)$ and the central obstruction $[\mathfrak{g}] \in H^2(T^2; Z(K))$ of a class-≤ 2 inventory gerbe are *not independent*: δ-coupling forces K nonabelian, and every nonabelian class-2 group has $[\mathfrak{g}] \neq 0$ (central extension non-split, trivial center-action), so $\delta \neq 0 \Rightarrow [\mathfrak{g}] \neq 0$ — a strict one-way forced correlation **(B)**; the converse fails at $K = D_4$ ($[\mathfrak{g}] \neq 0$, $\delta \equiv 0$, the dihedral $4 \mid n$ witness), and the corner $(\delta \neq 0, [\mathfrak{g}] = 0)$ is forced empty.